

CBSE Class 12 Chemistry — Half-Yearly Predicted Paper (Set 5 of 10)

50 Most-Probable PYQs & Board-Pattern Practice Questions

Section A: 1-Mark Questions (MCQs / Assertion-Reason)

Q1. The ebullioscopic constant of water is $0.52 \text{ K kg mol}^{-1}$. What is the approximate boiling point of a **1 molal** aqueous solution of a non-volatile, non-electrolyte solute? (a) 100°C (b) 100.52°C (c) 99.48°C (d) 0.52°C

[Solutions] [Practice] [Confidence: Very High]

Q2. The rusting of iron is fundamentally a process envisaged as the setting up of an: (a) Electrolytic cell (b) Electrochemical cell (c) Concentration cell (d) Irreversible fuel cell [Electrochemistry] [PYQ 2019] [Confidence: Very High]

Q3. The time required for the **99.9%** completion of a first-order reaction is approximately how many times its half-life ($t_{1/2}$)? (a) 2 times (b) 3.3 times (c) 10 times (d) 100 times [Chemical Kinetics] [PYQ 2015] [Confidence: High]

Q4. Mischmetall is a well-known alloy consisting predominantly of: (a) A lanthanoid metal and iron (b) Actinoid metals and copper (c) Titanium and steel (d) Magnesium and aluminum [The d- and f-Block Elements] [PYQ 2017] [Confidence: Very High]

Q5. Which of the following is a homoleptic complex? (a) $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$ (b) $[\text{Pt}(\text{en})_2\text{Cl}_2]^{2+}$ (c) $[\text{Co}(\text{NH}_3)_6]^{3+}$ (d) $[\text{Ni}(\text{NH}_3)_4(\text{H}_2\text{O})_2]^{2+}$ [Coordination Compounds] [CBSE Term-2 2022] [Confidence: Very High]

Q6. The reaction of ethyl bromide with alcoholic **AgCN** predominantly yields: (a) Ethyl cyanide (b) Ethyl isocyanide (c) Propane (d) Ethyl isocyanate [Haloalkanes and Haloarenes] [PYQ 2018] [Confidence: Very High]

Q7. Phenol, upon vigorous oxidation with chromic acid ($\text{Na}_2\text{Cr}_2\text{O}_7/\text{H}_2\text{SO}_4$), produces a conjugated diketone known as: (a) Benzoic acid (b) Salicylic acid (c) p-Benzoquinone (d) Benzophenone [Alcohols, Phenols and Ethers] [PYQ 2019] [Confidence: Very High]

Q8. The specific reagent used in the Etard reaction for the oxidation of toluene to benzaldehyde is: (a) CrO_2Cl_2 in CS_2 (b) KMnO_4 and H_2SO_4 (c) SnCl_2 and HCl (d) Alkaline KMnO_4 [Aldehydes, Ketones and Carboxylic Acids] [PYQ 2021] [Confidence: Very High]

Q9. According to Henry's law, the value of the constant K_H for a given gas-solvent system: (a) Increases with an increase in temperature (b) Decreases with an increase in temperature (c) Remains constant regardless of temperature (d) Depends strictly on the volume of the container [Solutions] [PYQ 2020] [Confidence: High]

Q10. The number of chiral carbon atoms present in the molecule of 2-chlorobutane is: (a) 0 (b) 1 (c) 2 (d) 3 [Haloalkanes and Haloarenes] [Practice] [Confidence: Very High]

Q11. By international convention, the standard electrode potential of the Standard Hydrogen Electrode (SHE) at all temperatures is assigned a value of exactly: (a) **1.00 V** (b) **0.00 V** (c) **10.0 V** (d) **−1.00 V** [Electrochemistry] [PYQ 2016] [Confidence: Very High]

Q12. What is the appropriate unit for the rate constant of a reaction having an overall order of 1.5? (a) $\text{mol}^{-1}\text{L s}^{-1}$ (b) $\text{mol}^{1/2}\text{L}^{-1/2}\text{s}^{-1}$ (c) $\text{mol}^{-1/2}\text{L}^{1/2}\text{s}^{-1}$ (d) s^{-1} [Chemical Kinetics] [Practice] [Confidence: Medium]

Directions for Q13 to Q16: A statement of Assertion (A) followed by a statement of Reason (R) is given. Choose the correct answer out of the following choices: (a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is not the correct explanation of A. (c) A is true but R is false. (d) A is false but R is true.

Q13. Assertion (A): The separation of Zirconium (**Zr**) and Hafnium (**Hf**) in their mineral state is extremely difficult. Reason (R): Due to lanthanoid contraction, **Zr** and **Hf** possess almost identical atomic and ionic radii. [The d- and f-Block Elements] [PYQ 2018] [Confidence: Very High]

Q14. Assertion (A): Alcohols can behave both as Bronsted acids as well as Bronsted bases. Reason (R): Alcohols can donate a proton to form an alkoxide ion (acidic nature), and the unshared electron pairs on oxygen make them proton acceptors (basic nature). [Alcohols, Phenols and Ethers] [PYQ 2020] [Confidence: High]

Q15. Assertion (A): The order of a chemical reaction can be a fractional number. Reason (R): The molecularity of an elementary reaction can be zero. [Chemical Kinetics] [PYQ 2017] [Confidence: Very High]

Q16. Assertion (A): Electrophilic substitution reactions in benzoic acid typically take place at the meta position. Reason (R): The carboxyl group ($-\text{COOH}$) is strongly electron-withdrawing and deactivates the benzene ring, making the meta position relatively more electron-rich than ortho/para positions. [Aldehydes, Ketones and Carboxylic Acids] [PYQ 2015] [Confidence: Very High]

Section B: 2-Mark Questions (Short Answer)

Q17. Define the term *isotonic solutions*. What happens specifically to human red blood cells (which are isotonic with **0.9% NaCl** solution) when they are placed in a **1.2% NaCl** solution? [Solutions] [PYQ 2016] [Confidence: Very High]

Q18. Write the exact IUPAC name of the coordination complex $[\text{Pt}(\text{NH}_3)_2\text{Cl}(\text{NH}_2\text{CH}_3)]\text{Cl}$ and specify the coordination number of the central metal atom. [Coordination Compounds] [PYQ 2018] [Confidence: High]

Q19. Give a chemical reason explaining why a standard dry cell (Leclanché cell) becomes completely dead after a long time even if it has not been used. [Electrochemistry] [PYQ 2014] [Confidence: Very High]

Q20. Name a transition element from the 3d series that strictly does not exhibit variable oxidation states. Give one general reason why transition metals frequently form complex compounds. [The d- and f-Block Elements] [PYQ 2017] [Confidence: Very High]

Q21. How will you seamlessly convert Phenol to Aspirin (2-acetoxybenzoic acid) in organic chemistry? Write the required chemical reactions. [Alcohols, Phenols and Ethers] [PYQ 2019] [Confidence: Very High]

Section C: 3-Mark Questions (Short Answer)

Q22. Calculate the exact mass of a non-volatile solute (Molar mass = **40 g mol⁻¹**) which should be dissolved in **114 g** of octane to reduce its vapour pressure to precisely **80%** of its original value. (Given: Molar mass of octane = **114 g mol⁻¹**). [Solutions] [PYQ 2020] [Confidence: Very High]

Q23. Show mathematically that the time required for the **99%** completion of a first-order chemical reaction is exactly twice the time required for its **90%** completion. [Chemical Kinetics] [PYQ 2018] [Confidence: Very High]

Q24. Calculate the e.m.f. of the following cell at **298 K**:

$\text{Sn}(s)|\text{Sn}^{2+}(0.004\text{ M})||\text{H}^+(0.020\text{ M})|\text{H}_2(g)(1\text{ bar})|\text{Pt}(s)$ Given: $E^\circ_{\text{Sn}^{2+}/\text{Sn}} = -0.14\text{ V}$. [Electrochemistry] [PYQ 2018] [Confidence: High]

Q25. Assign suitable chemical reasons for the following observations: (i) Transition metals show their highest oxidation states exclusively in their oxides and fluorides. (ii) The standard electrode potential $E^\circ(\text{Cu}^{2+}/\text{Cu})$ for copper is notably positive (+0.34 V). (iii) The Actinoid contraction is greater from element to element than the Lanthanoid contraction. [The d- and f-Block Elements] [PYQ 2023 / 2020] [Confidence: Very High]

Q26. Using Valence Bond Theory (VBT), clearly deduce the state of hybridization, the resulting geometry, and the magnetic behaviour of the complex ion $[\text{Co}(\text{C}_2\text{O}_4)_3]^{3-}$. (Given: Atomic number of Cobalt = 27). [Coordination Compounds] [CBSE Term-2 2022] [Confidence: Very High]

Q27. Give specific chemical reasons for the following facts: (i) Haloalkanes react with aqueous **KCN** to predominantly form alkyl cyanides, but react with **AgCN** to predominantly form isocyanides. (ii) Electrophilic substitution reactions in haloarenes occur at a much slower rate compared to normal benzene. (iii) A racemic mixture is completely optically inactive despite containing optically active enantiomers. [Haloalkanes and Haloarenes] [PYQ 2019 / 2020] [Confidence: Very High]

Q28. Write the proper chemical equations to show how you will efficiently carry out the following conversions: (i) Benzene to m-nitroacetophenone (ii) Propanone to propene (iii) Benzoic acid to benzamide [Aldehydes, Ketones and Carboxylic Acids] [PYQ 2015 / Practice] [Confidence: High]

Section D: 4-Mark Questions (Case-Study)

Q29. Case Study: Stereochemistry of Nucleophilic Substitution Alkyl halides readily undergo nucleophilic substitution reactions. The stereochemical outcomes of these reactions are incredibly important to understand the underlying mechanisms. A generic S_N2 reaction proceeds via a concerted transition state and results in the complete inversion of configuration (Walden inversion). Conversely, an S_N1 reaction proceeds via a planar carbocation intermediate. When an optically active alkyl halide undergoes an S_N1 reaction, the nucleophile can attack the planar intermediate from both faces, leading to racemization. (a) Out of 1-chlorobutane and 2-chlorobutane, identify the chiral molecule. (1 Mark) (b) What is exactly meant by the term "retention of configuration" in stereochemistry? (1 Mark) (c) An optically active alkyl halide 'A' with the formula C_4H_9Br undergoes an S_N1 reaction with aqueous KOH . Draw the exact structure of compound 'A' and briefly state the stereochemical outcome of this specific reaction in the product. (2 Marks) [Haloalkanes and Haloarenes] [CBSE SQP 2022 Themes] [Confidence: Very High]

Q30. Case Study: Corrosion as an Electrochemical Process Corrosion is a massive industrial problem that is fundamentally an electrochemical process. The rusting of iron happens in the presence of water and air. At a particular spot on an iron object, oxidation takes place and that spot behaves as a miniature anode. The electrons released at this anodic spot move through the bulk metal to another spot where they reduce atmospheric oxygen in the presence of H^+ ions, effectively acting as a miniature cathode. (a) Write the specific anodic and cathodic half-cell chemical reactions occurring during the rusting of iron. (2 Marks) (b) Write the overall redox reaction of the rusting process before the formation of rust crystals. (1 Mark) (c) Name any two practical methods extensively used in industry to prevent the corrosion of iron. (1 Mark) [Electrochemistry] [PYQ 2014 / Practice] [Confidence: High]

Section E: 5-Mark Questions (Long Answer)

Q31. (a) The rate constant for a reaction of zero order in reactant A is $0.0030 \text{ mol L}^{-1} \text{ s}^{-1}$. Exactly how long will it take for the initial concentration of A to fall from 0.10 M to 0.075 M ? (2 Marks) (b) Define the term "Collision frequency" (Z) in the context of the collision theory of chemical kinetics. (1 Mark) (c) Describe the specific effect of adding a positive catalyst on the activation energy (E_a) of a reaction. Draw a labelled potential energy profile curve comparing the catalysed and uncatalysed pathways for an exothermic reaction. (2 Marks) [Chemical Kinetics] [PYQ 2018 / 2020] [Confidence: Very High]

Q32. (a) Write the detailed mechanism (using proper curved arrows) of the acid-catalysed hydration of ethene to yield ethanol. (3 Marks) (b) How will you synthesise the following ethers strictly using Williamson's ether synthesis? Choose the correct alkyl halides and sodium alkoxides and write the equations. (i) 1-Propoxypropane (ii) 2-Methoxy-2-methylpropane (tert-butyl methyl ether) (2 Marks) [Alcohols, Phenols and Ethers] [PYQ 2019 / 2015] [Confidence: Very High]

Q33. (a) An organic compound 'A' with the molecular formula $C_5H_{10}O$ forms an addition product with sodium bisulphite and gives a positive iodoform test (yellow precipitate). However, it does not reduce Tollen's reagent or Fehling's solution. Compound 'A' on reduction with sodium borohydride ($NaBH_4$) gives pentan-2-ol. Identify the exact structure and IUPAC name of compound 'A' and write the chemical reactions for its iodoform and bisulphite tests. (3 Marks) (b) Give reliable, simple chemical tests to uniquely distinguish between the following pairs of compounds: (i) Pentan-2-one and Pentan-3-one (ii) Benzaldehyde and Acetophenone (2 Marks) [Aldehydes, Ketones and Carboxylic Acids] [PYQ 2014 / 2016] [Confidence: Very High]

Answer Key

Q1. (b) 100.52°C . ($\Delta T_b = K_b \cdot m = 0.52 \times 1 = 0.52$. Boiling point = $100 + 0.52$). **Q2.** (b) Electrochemical cell. **Q3.** (c) 10 times. ($t_{99.9} = \frac{2.303}{k} \log(10^3) = \frac{6.909}{k}$. $t_{1/2} = \frac{0.693}{k}$. Ratio ≈ 10). **Q4.** (a) A lanthanoid metal and iron. (approx 95% lanthanoid, 5% iron, traces of S, C). **Q5.** (c) $[\text{Co}(\text{NH}_3)_6]^{3+}$. (Contains only one kind of donor group/ligand). **Q6.** (b) Ethyl isocyanide. **Q7.** (c) p-Benzoquinone. **Q8.** (a) CrO_2Cl_2 in CS_2 . **Q9.** (a) Increases with an increase in temperature. (Thus, solubility decreases at higher temps). **Q10.** (b) 1. (Carbon-2 is bonded to 4 different groups: H, Cl, Methyl, Ethyl). **Q11.** (b) 0.00 V . **Q12.** (c) $\text{mol}^{-1/2}\text{L}^{1/2}\text{s}^{-1}$. (Unit is $(\text{mol/L})^{1-n}\text{s}^{-1}$. Here $n = 1.5$, so $(\text{mol/L})^{-0.5}\text{s}^{-1}$). **Q13.** (a) Both A and R are true and R is the correct explanation of A. **Q14.** (a) Both A and R are true and R is the correct explanation of A. **Q15.** (c) A is true but R is false. (Molecularity cannot be zero). **Q16.** (a) Both A and R are true and R is the correct explanation of A.

Q17. Isotonic solutions have the exact same osmotic pressure at a given temperature. When red blood cells are placed in 1.2% NaCl (a hypertonic solution), water flows out of the cells via exosmosis, causing the cells to shrink (plasmolysis). **Q18.** Diamminechlorido(methanamine)platinum(II) chloride. Coordination number = 4. **Q19.** Even when not in use, the highly acidic paste of NH_4Cl slowly and continuously corrodes the zinc metal container (anode), eventually rendering the cell dead. **Q20.** Scandium (Sc) exhibits only a +3 oxidation state. Transition metals form complexes due to their small ionic size, high nuclear charge, and the availability of vacant d-orbitals to accept lone pairs from ligands. **Q21.** Step 1 (Kolbe's): Phenol reacts with NaOH to form sodium phenoxide, which reacts with CO_2 at 400 K followed by acidification to yield Salicylic acid (2-hydroxybenzoic acid). Step 2: Salicylic acid is acetylated using Acetic anhydride in the presence of an acid catalyst (H^+) to form Aspirin.

Q22. Relative lowering of VP: $\frac{P^\circ - P}{P^\circ} = \frac{n_B}{n_A + n_B} \approx X_B$. Given P reduces to 80% , so $\frac{P^\circ - 0.8P^\circ}{P^\circ} = 0.20 = \frac{1}{5}$. Moles of octane (n_A) = $\frac{114}{114} = 1\text{ mol}$. $X_B = \frac{n_B}{n_B + 1} = \frac{1}{5} \Rightarrow 5n_B = n_B + 1 \Rightarrow 4n_B = 1 \Rightarrow n_B = 0.25\text{ mol}$.

Mass of solute = $n_B \times \text{Molar mass} = 0.25 \times 40 = 10\text{ g}$. **Q23.** For a 1st order reaction, $t = \frac{2.303}{k} \log \left(\frac{[R]_0}{[R]} \right)$.

$t_{99\%} = \frac{2.303}{k} \log \left(\frac{100}{1} \right) = \frac{2.303}{k} \times 2$. $t_{90\%} = \frac{2.303}{k} \log \left(\frac{100}{10} \right) = \frac{2.303}{k} \log(10) = \frac{2.303}{k} \times 1$. Therefore, $t_{99\%} = 2 \times t_{90\%}$. **Q24.** Overall reaction: $\text{Sn} + 2\text{H}^+ \rightarrow \text{Sn}^{2+} + \text{H}_2$. Here, $n = 2$.

$E^\circ_{\text{cell}} = E^\circ_{\text{cathode}} - E^\circ_{\text{anode}} = 0.00 - (-0.14) = 0.14\text{ V}$.

$E_{\text{cell}} = E^\circ_{\text{cell}} - \frac{0.0591}{2} \log \frac{[\text{Sn}^{2+}]}{[\text{H}^+]^2} = 0.14 - 0.0295 \log \frac{0.004}{(0.020)^2} \cdot \frac{0.004}{0.0004} = 10$.

$E_{\text{cell}} = 0.14 - 0.0295 \log(10) = 0.14 - 0.0295 = 0.1105\text{ V}$. **Q25.** (i) Oxygen and fluorine are highly electronegative and very small in size, enabling them to completely oxidize the metal. (ii) Copper has high enthalpy of atomization and ionization enthalpies which are not fully compensated by its hydration enthalpy. (iii) The 5f electrons in actinoids have a much poorer shielding effect than the 4f electrons in lanthanoids, causing a stronger inward pull by the nucleus. **Q26.** Co is $3d^7 4s^2$. Co^{3+} is $3d^6$. Oxalate ($\text{C}_2\text{O}_4^{2-}$) is a strong field bidentate ligand for Co^{3+} , causing all 6 d-electrons to pair up into the t_{2g} level. This leaves two 3d orbitals empty. Hybridisation is d^2sp^3 (Inner orbital complex). Geometry is Octahedral. Magnetic property is purely Diamagnetic (0 unpaired electrons). **Q27.** (i) KCN is ionic, yielding ambient CN^- where attack via the less electronegative Carbon forms a stronger C-C bond (cyanide). AgCN is largely covalent, leaving only Nitrogen's lone pair available for attack, forming isocyanide. (ii) Halogens pull electron density via the strong -I effect, deactivating the benzene ring toward electrophilic attack. (iii) It contains equal proportions of both dextro (+) and laevo (-) enantiomers; the optical rotation of one perfectly cancels the rotation of the other. **Q28.** (i) Benzene $\xrightarrow{\text{CH}_3\text{COCl}/\text{Anhyd. AlCl}_3}$ Acetophenone $\xrightarrow{\text{Conc. HNO}_3/\text{Conc. H}_2\text{SO}_4}$ m-Nitroacetophenone. (ii) Propanone $\xrightarrow{\text{NaBH}_4}$ Propan-2-ol $\xrightarrow{\text{Conc. H}_2\text{SO}_4, 443\text{K}}$ Propene. (iii) Benzoic acid $\xrightarrow{\text{SOCl}_2}$ Benzoyl chloride $\xrightarrow{\text{NH}_3}$ Benzamide. (Or directly treat with NH_3 to form ammonium benzoate, then heat vigorously).

Q29. (a) 2-Chlorobutane. (b) The preservation of the spatial arrangement of bonds to an asymmetric centre during a chemical reaction or transformation. (c) Structure: $\text{CH}_3\text{CH}_2\text{CH}(\text{Br})\text{CH}_3$ (2-bromobutane). Stereochemical outcome: Racemization (formation of a 50:50 racemic mixture of (+)-butan-2-ol and (-)-butan-2-ol) because the planar carbocation intermediate is attacked from both faces equally. **Q30.** (a) Anode: $2\text{Fe}(s) \rightarrow 2\text{Fe}^{2+}(aq) + 4e^-$. Cathode: $\text{O}_2(g) + 4\text{H}^+(aq) + 4e^- \rightarrow 2\text{H}_2\text{O}(l)$. (b) Overall:

$2Fe(s) + O_2(g) + 4H^+(aq) \rightarrow 2Fe^{2+}(aq) + 2H_2O(l)$. (c) Galvanization (coating with zinc) and Cathodic protection (sacrificial anodes like Mg).

Q31. (a) For a zero-order reaction: $[A] = [A]_0 - kt \Rightarrow t = \frac{[A]_0 - [A]}{k}$. $t = \frac{0.10 - 0.075}{0.0030} = \frac{0.025}{0.0030} = 8.33 \text{ seconds}$.

(b) The number of collisions per second per unit volume of the reaction mixture. (c) A positive catalyst chemically lowers the activation energy (E_a) by providing an entirely alternative reaction pathway or mechanism. (Draw an Exothermic Potential Energy vs Reaction Coordinate graph. Draw a high peak for uncatalysed path and a noticeably lower peak for the catalysed path, both starting at reactants and ending lower at products). **Q32.** (a)

Step 1: Protonation of alkene to form a carbocation: $CH_2 = CH_2 + H_3O^+ \rightleftharpoons CH_3 - CH_2^+ + H_2O$. Step 2: Nucleophilic attack of water on the carbocation: $CH_3 - CH_2^+ + H_2O \rightleftharpoons CH_3 - CH_2 - O^+H_2$. Step 3: Deprotonation to form ethanol: $CH_3 - CH_2 - O^+H_2 + H_2O \rightleftharpoons CH_3 - CH_2 - OH + H_3O^+$. (b) (i) 1-Propoxypropane: React 1-Bromopropane with Sodium propoxide.

$CH_3CH_2CH_2Br + CH_3CH_2CH_2ONa \rightarrow CH_3CH_2CH_2 - O - CH_2CH_2CH_3 + NaBr$. (ii) 2-Methoxy-2-methylpropane: Must react Bromomethane (primary halide) with Sodium tert-butoxide (bulky base).

$CH_3Br + (CH_3)_3CONa \rightarrow (CH_3)_3C - O - CH_3 + NaBr$. (Reversing these yields an alkene via elimination). **Q33.** (a) A is Pentan-2-one ($CH_3 - CO - CH_2CH_2CH_3$). Iodoform reaction:

$CH_3COCH_2CH_2CH_3 + 3I_2 + 4NaOH \rightarrow CHI_3 \downarrow (\text{yellow}) + CH_3CH_2CH_2COONa + 3NaI + 3H_2O$

. Bisulphite reaction: $CH_3COCH_2CH_2CH_3 + NaHSO_3 \rightleftharpoons \text{Bisulphite adduct (solid crystals)}$. (b) (i)

Iodoform test: Pentan-2-one gives a yellow precipitate of CHI_3 with $I_2/NaOH$. Pentan-3-one does not (lacks terminal methyl ketone). (ii) Iodoform test: Acetophenone ($PhCOCH_3$) gives a yellow precipitate with $I_2/NaOH$. Benzaldehyde ($PhCHO$) does not.

CBSE Class 12 (2027) — Half-Yearly Predicted Paper (80% syllabus). AI-generated via PYQ/Blueprint analysis, not actual exam questions.